

Exact Banach Support-Sum Criteria for Tao-Type Uncertainty Principles

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Source and Open Problem

The source paper is K. Mahesh Krishna, *Functional Donoho-Stark-Elad-Bruckstein-Ricaud-Torresani Uncertainty Principle*, arXiv:2304.03324. In Section 3, after recalling Tao’s prime-dimensional uncertainty principle for the Fourier transform on $\mathbb{C}^p$, the paper asks:

“What is the version Tao uncertainty principle for Banach spaces of prime dimension?”

It is well-known that Donoho-Stark uncertainty principle cannot be improved [4]. However, Donoho-Stark uncertainty principle gives (using AM-GM inequality) that

$$(4) \quad \|h\|_0 + \|\widehat{h}\|_0 \geq 2\sqrt{d}, \quad \forall h \in \mathbb{C}^d \setminus \{0\}, \forall d \in \mathbb{N}.$$

In 2005, Tao improved Inequality (4) in prime dimensions [16].

Theorem 3.1. (*Tao Uncertainty Principle*) [16] *Let $p \in \mathbb{N}$ be a prime number. Then*

$$\|h\|_0 + \|\widehat{h}\|_0 \geq p + 1 \geq 2\sqrt{p}, \quad \forall h \in \mathbb{C}^p \setminus \{0\}.$$

Motivated from Tao uncertainty principle, we ask the following problem.

Problem 3.2. *What is the version Tao uncertainty principle for Banach spaces of prime dimension?*

REFERENCES

- [1] P. G. Casazza, S. J. Dilworth, E. Odell, Th. Schlumprecht, and A. Zsák. Coefficient quantization for frames in Banach spaces. *J. Math. Anal. Appl.*, 348(1):66–86, 2008.

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Result

For arbitrary finite p -Schauder frames, every support-sum lower bound is governed by a deletion/spanning condition on the remaining analysis functionals. In the basis case this becomes an especially simple full-spark criterion: the Tao support-sum bound is equivalent to the transition matrix being full spark. Thus the Banach-space replacement for the Fourier/primal mechanism is not prime dimension by itself, but the full-spark property of the two coordinate systems.

Setup

Let $\mathcal{X}$ be a d -dimensional Banach space over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Let

$$(\{f_j\}_{j=1}^n, \{\tau_j\}_{j=1}^n), \quad (\{g_k\}_{k=1}^m, \{\omega_k\}_{k=1}^m)$$

be p-Schauder frames in the sense of the source paper. Only their analysis functionals are needed below. The analysis maps

$$\theta_f x = (f_j(x))_{j=1}^n, \quad \theta_g x = (g_k(x))_{k=1}^m$$

are injective; for p-Schauder frames they are isometries.

For $A \subseteq \{1, \dots, n\}$ and $B \subseteq \{1, \dots, m\}$ define

$$E(A, B) = \bigcap_{j \notin A} \ker f_j \cap \bigcap_{k \notin B} \ker g_k.$$

Equivalently, $E(A, B)$ is the common kernel of all analysis functionals that remain after deleting the coordinates in A and B .

Proof Intuition

For general p-Schauder frames, a vector has small support exactly when it lies in many coordinate kernels. Thus a lower bound on

$$|\text{supp}(\theta_f x)| + |\text{supp}(\theta_g x)|$$

is equivalent to saying that no nonzero vector can survive after all but a small number of coordinates are forced to vanish. In the basis case this deletion condition is the familiar full-spark condition on the transition matrix. Tao's finite cyclic prime theorem rests on the corresponding algebraic fact for the prime Fourier matrix: every square minor is nonzero.

Theorem 1 (Exact frame support-sum criterion). *Let $L \geq 1$ be an integer. With the notation above, the following are equivalent.*

1. *For every $x \in \mathcal{X} \setminus \{0\}$,*

$$|\text{supp}(\theta_f x)| + |\text{supp}(\theta_g x)| \geq L.$$

2. *For every $A \subseteq \{1, \dots, n\}$ and $B \subseteq \{1, \dots, m\}$ with*

$$|A| + |B| < L,$$

one has $E(A, B) = \{0\}$.

3. *Equivalently, for every such A, B , the family*

$$\{f_j : j \notin A\} \cup \{g_k : k \notin B\}$$

spans $\mathcal{X}^$.*

Proof. For $x \neq 0$, put

$$A = \text{supp}(\theta_f x), \quad B = \text{supp}(\theta_g x).$$

Then $x \in E(A, B)$. Therefore if $|A| + |B| < L$ and $E(A, B) = \{0\}$ for all such pairs, no nonzero x can have support sum below L .

Conversely, if $E(A, B)$ contains a nonzero vector x for some pair with $|A| + |B| < L$, then all nonzero coordinates of $\theta_f x$ lie in A and all nonzero coordinates of $\theta_g x$ lie in B . Hence

$$|\text{supp}(\theta_f x)| + |\text{supp}(\theta_g x)| \leq |A| + |B| < L,$$

contradicting the support-sum bound.

Finally, a family of linear functionals spans $\mathcal{X}^*$ if and only if its common kernel is zero, giving the equivalence of (2) and (3). $\square$

Corollary 1 (Tao-type frame criterion). *For arbitrary p -Schauder frames on a d -dimensional Banach space, the support-sum bound*

$$|\text{supp}(\theta_f x)| + |\text{supp}(\theta_g x)| \geq d + 1, \quad x \neq 0,$$

holds if and only if, after deleting any total of at most d coordinates from the two analysis families, the remaining functionals still span $\mathcal{X}^$.*

Basis Specialization

Now suppose $n = m = d$ and the two frames are bases with biorthogonal coordinate functionals. Write

$$A = (g_k(\tau_j))_{k,j=1}^d.$$

Then $(g_k(x)) = A(f_j(x))$. Call A *full spark* if every square minor of A is nonzero.

Theorem 2 (Basis full-spark criterion). *For two bases of $\mathcal{X}$, the following are equivalent.*

1. *For every $x \in \mathcal{X} \setminus \{0\}$,*

$$|\text{supp}(f_j(x))| + |\text{supp}(g_k(x))| \geq d + 1.$$

2. *The transition matrix $A = (g_k(\tau_j))_{k,j=1}^d$ is full spark.*

Proof. This follows from the preceding corollary, but we spell out the standard matrix form. If A is not full spark, choose $R, S \subseteq \{1, \dots, d\}$ with $|R| = |S| = s$ such that $A_{R,S}$ is singular. There is a nonzero coordinate vector c supported in S with $A_{R,S} c_S = 0$. For $x = \sum_j c_j \tau_j$ one has

$$|\text{supp}(f_j(x))| \leq s, \quad |\text{supp}(g_k(x))| \leq d - s,$$

so the support sum is at most d .

Conversely, suppose there is a nonzero x with support sets

$$S = \text{supp}(f_j(x)), \quad T = \text{supp}(g_k(x)).$$

If $s + t = |S| + |T| \leq d$, then every row in $R_0 = \{1, \dots, d\} \setminus T$ annihilates the nonzero vector $c_S = (f_j(x))_{j \in S}$ through $A_{R_0,S}$. Since $|R_0| = d - t \geq s = |S|$, choose $R \subseteq R_0$ with $|R| = s$. Then

$$A_{R,S} c_S = 0$$

with $c_S \neq 0$, so the square minor $A_{R,S}$ is singular. Therefore A is not full spark. $\square$

Corollary 2 (Prime-dimensional Banach formulation). *Let $r > 1$ be prime. For two p -orthonormal bases in an r -dimensional Banach space, the Tao-type bound*

$$|\text{supp}(\theta_f x)| + |\text{supp}(\theta_g x)| \geq r + 1, \quad x \neq 0,$$

holds if and only if their transition matrix $(g_k(\tau_j))$ is full spark.

Proof. Apply the theorem with $d = r$. The proof does not use primality; in the classical Fourier case, primality enters through Tao's theorem that the prime Fourier matrix is full spark. $\square$

Corollary 3 (Prime dimension alone is insufficient). *There is no dimension-only Tao principle for arbitrary Banach p -orthonormal bases. For every prime $r > 1$ and every $1 < p < \infty$, take $\mathcal{X} = \ell_p^r$ and take the same canonical p -orthonormal basis twice. For $x = e_1$,*

$$|\text{supp}(\theta_f x)| + |\text{supp}(\theta_g x)| = 1 + 1 = 2 < r + 1.$$

Scope and Limitations

This is a strong partial answer to the source problem because Problem 3.2 asks broadly for a “version” of Tao’s theorem, not for a single formal conjecture. The packet settles the natural support-sum formulation for arbitrary finite p -Schauder frames and gives the clean full-spark specialization for p -orthonormal bases. It also shows that the prime-dimensional hypothesis alone cannot replace the Fourier full-spark mechanism.

The result does not claim to choose a unique canonical Banach-space analogue among possible entropy, product, weighted-support, or recovery-theoretic Tao variants. It gives the exact answer for the support-sum interpretation suggested by the source’s use of support cardinalities.

Novelty and Literature Check

The run indexes were searched for 2304.03324, the source title, Tao uncertainty terminology, prime dimension, and full spark. No duplicate packet was found. Web searches on June 28, 2026 for the exact Problem 3.2 wording, “Functional Tao uncertainty principle”, and “Tao uncertainty principle Banach spaces prime dimension” surfaced the source paper and adjacent Krishna followups but no later explicit answer to Problem 3.2.

Human Review Recommendation

Review as a likely valid strong partial result. The key verification is the elementary equivalence between support-sum lower bounds and the deletion/spanning condition. The full-spark basis corollary and the warning that prime dimension alone is insufficient then follow immediately.

References

- [1] K. Mahesh Krishna, *Functional Donoho-Stark-Elad-Bruckstein-Ricaud-Torresani Uncertainty Principle*, arXiv:2304.03324, 2023.
- [2] Terence Tao, *An uncertainty principle for cyclic groups of prime order*, Math. Res. Lett. 12 (2005), no. 1, 121–127.